

Using SQLite in Android

The Android operating system comes with a built-in SQLite DBMS, allowing users to save data like contacts and text messages. This article explains how to retrieve and save values in the embedded database.



The Android architecture, which most developers must be familiar with, consists of a *Libraries* layer, which in turn consists of SQLite, an open source database management system that is used to create databases and tables on the Android device (Figure 1).

SQLite supports various datatypes such as *int*, *varchar*, *blob*, *real*, *double*, *float*, *text*, *Boolean*, *date*, *datetime*, etc.

Let us look at how to store values in the embedded database and retrieve them through an Android app. In order to use the database, import the SQLite package as follows:

```
import android.database.sqlite;
```

Your class should extend the `SQLiteOpenHelper` class in order to use the methods to insert and retrieve values into and from the database. When extending this class, override the constructor and pass the required values.

```
public class MyDatabaseHelper extends SQLiteOpenHelper {
    public MyDatabaseHelper (Context context,
        String dbName, SQLiteDatabase.CursorFactory cursorFactory,
        int version) {
        super(context, dbName, null,
    };
}
```

This method takes as its parameters the context that is used to open or create the database, the name of the database file, the cursor factory which indicates the cursor behaviour, and the version number of the database. The version starts from 1.

Instantiate the helper class from your activity

To instantiate, type:

```
MyDatabaseHelper helper = new MyDatabaseHelper(getApplication
Context(), "osfy.db", null, 1);
```

The null value of `CursorFactory` indicates default behavior. Use the `execSQL` method in the helper class to execute SQL queries.

```
public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE Article (id INT, title VARCHAR(50),
author (50))");
}
```

The same method can be used to insert values.

```
public void insertValues() {
    SQLiteDatabase db = this.
getWritableDatabase();
    db.execSQL("INSERT INTO Article VALUES(1, 'Title of the
article', 'Name of the author')");
}
```

If you are retrieving values from `EditText` views in the app, use the `getText` method and replace the parameters with the values entered by the user.

In order to retrieve values from the database, use the `rawQuery` method, as follows:

```
public void getData() {
    SQLiteDatabase db = this.
getReadableDatabase();
    Cursor result = db.rawQuery("select * from
Article", null);
```

Continued on Page 75...